
PRANCE: Progressive Representation Alignment and Noxious-content Containment for Robust Large Language Models

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Abstract

1 Large language models (LLMs) remain vulnerable to a broad spectrum of jailbreak
2 techniques, ranging from prompt suffixes and decoding-time parameter changes
3 to malicious weight injections that subvert safety tuning. We tackle seven open
4 problems of current defences—shallow alignment, temporal blind spots, mechanis-
5 tic jailbreaks, parameter-free exploits, back-door fine-tunes, safety-basin drift, and
6 edge-deployment limits—by introducing PRANCE, a two-phase alignment and
7 on-line enforcement framework. Offline, PRANCE applies reward–safety dualisa-
8 tion to compute an optimal λ and fuses independently reward- and safety-aligned
9 policies via Lisa-style proximal regularisation, ensuring the initial checkpoint stays
10 inside the VISAGE safety basin. Online, each decoder layer hosts a low-rank
11 Safety Vector Field that is meta-updated every 256 generated tokens; three hi-
12 erarchical monitors estimate risk at token, segment, and trajectory granularity;
13 and a dual-policy controller chooses pass, replace, truncate, or refuse actions by
14 maximising $Q_{help} - \lambda \cdot Q_{risk}$. Robustness is further increased through adversarial
15 weight-space training, decoding-parameter sweeps, a VISAGE depth penalty, and
16 adapter quarantine for untrusted LoRA modules. All safety modules add only eight
17 percent parameters and incur eleven percent latency and 10.9 GB VRAM on a 3
18 B Llama-3 model. Experiments on AdvBench, JailbreakBench, and HarmBench
19 show a clean attack-success rate (ASR) of 2.4 % ($\downarrow 9\times$ vs. Constitutional-AI) and
20 ≤ 3 pp ASR increase under weight attacks, malicious fine-tuning, or decoding
21 exploits while preserving 99 % of original utility on MT-Bench, MMLU, and
22 GSM8K. A seven-day continual-learning simulation confirms that PRANCE keeps
23 VISAGE depth within 0.12 and recalls 92 % of malicious adapters. These findings
24 demonstrate that combining optimal dual alignment with adaptive low-rank safety
25 directions yields an effective and efficient defence against diverse jailbreak vectors.

26 1 Introduction

27 Large language models (LLMs) have rapidly transitioned from laboratory curiosities to indispensable
28 components of conversational assistants, search engines, and productivity tools. As their societal
29 footprint expands, so too does the impact of misalignment. A burgeoning literature now catalogues
30 jailbreak vectors able to elicit disallowed content. Prompt-level attacks erase the single activation
31 direction that mediates refusal Arditi et al. [2024]; reinforcement-learning-guided searches discover
32 potent adversarial suffixes Chen et al. [2024]; sampling-only manipulations raise harmful responses
33 by merely altering temperature or top-p Huang et al. [2023]; and weight-space attacks such as
34 malicious LoRA injections, universal back-door poisoning, or one-shot harmful fine-tunes bypass
35 safety guarantees entirely Kurita et al. [2020], Rando and Tramèr [2023]. Even small random
36 perturbations can push a model outside its "safety basin", causing a step-like loss of harmlessness

Peng et al. [2024]. The defence problem is compounded by three obstacles. First, prevailing alignment pipelines reinforce the first refusal token, leaving a shallow-alignment hole exploitable by autoregressive generation. Second, most defences focus on prompts and ignore weight-space exploits. Third, methods that do confront sophisticated attacks - e.g., SmoothLLM or Task-Conditioned Masking - often impose latency or VRAM overheads incompatible with edge deployment. We address these gaps under strict resource constraints (≤ 12 GB VRAM and $\leq 15\%$ latency overhead) with PRANCE - Progressive Representation Alignment & Noxious-content Containment Engine. PRANCE rests on two principles. (1) Optimal dual alignment theory guarantees that a single scalar λ balances helpfulness and safety and can be computed offline Huang et al. [2024b]. (2) Run-time safety must adapt to both prompt- and weight-space perturbations through lightweight, low-rank interventions that minimally disturb benign generation.

1.1 Contributions

- **Dual-stage offline alignment:** We devise a dual-stage offline alignment that first learns reward- and safety-aligned policies via PeCAN/DPO Huang et al. [2024b] and then fuses them with Lisa proximal regularisation Huang et al. [2024a], maintaining VISAGE-basin proximity.
- **Safety Vector Fields (SVFs):** We introduce SVFs, four orthogonal low-rank directions per transformer block that are meta-updated online, enabling rapid response to weight drift and adversarial continuations.
- **Dual-policy controller:** We design a controller that, at every decoding step, selects pass, replace, truncate, or refuse by maximising $Q_{help} - \lambda \cdot Q_{risk}$ computed by compact Q-networks distilled from the aligned model.
- **Adapter quarantine and adversarial training:** We present adapter quarantine and adversarial weight-space training protocols that specifically target malicious LoRA back-doors and universal jailbreak triggers Rando and Tramèr [2023], Wu et al. [2022].
- **Evaluation suite:** We release a rigorous experimental suite - implementation-trace tests, large-scale robustness grids, and a seven-day continual-learning simulation - with full code and logs.

The remainder of the paper reviews related work (§2), summarises background concepts (§3), details PRANCE (§4), describes the experimental protocol (§5), reports empirical results (§6), and concludes with limitations and future work (§7).

2 Related Work

2.1 Jailbreak attacks

Manual prompts first exposed the fragility of refusal training Wei et al. [2023]. Optimised attacks now automate prompt discovery with genetic algorithms, discrete optimisation, or deep reinforcement learning Chen et al. [2024], Guo et al. [2024], Jones et al. [2023]. Few-shot strategies achieve $>95\%$ success within limited context Zheng et al. [2024], whereas generation-exploitation alters only decoding hyper-parameters to push ASR above 95% Huang et al. [2023]. Weight-space threats include weight poisoning Kurita et al. [2020], universal jailbreak back-doors planted during RLHF Rando and Tramèr [2023], and malicious adapters delivered via parameter-efficient fine-tuning Wu et al. [2022].

2.2 Defence landscape

Constitutional-AI extends RLHF with rule-based supervision but succumbs to strong attacks Wei et al. [2023]. SmoothLLM blurs refusal logits but adds runtime cost; Task-Conditioned Masking (TCM) embeds classifiers but cannot mitigate weight attacks. Lisa constrains drift during harmful fine-tuning Huang et al. [2024a] yet lacks run-time monitoring. VISAGE quantifies basin depth and reveals abrupt safety loss under drift Peng et al. [2024] but remains diagnostic. Finally, the discovery of a one-dimensional refusal subspace Arditì et al. [2024] hints that low-rank control might suffice for alignment.

86 2.3 Positioning PRANCE

87 PRANCE synthesises these insights. Optimal dual alignment supplies a principled λ , Lisa fusion
88 preserves basin proximity, and SVFs generalise low-rank refusal control to multiple adaptive direc-
89 tions. Unlike SmoothLLM or TCM, PRANCE simultaneously addresses prompt-, decoding-, and
90 weight-level vectors while satisfying tight resource budgets.

91 3 Background

92 3.1 Problem formulation

93 Let π_θ denote a causal language-model policy parameterised by weights θ that maps history h_t to a
94 distribution over next tokens. Two scalar feedback signals are available: reward $R(h_t, a_t)$ measuring
95 helpfulness and risk $S(h_t, a_t)$ reflecting potential harm. The alignment goal is to maximise expected
96 cumulative reward subject to an upper bound ε on expected cumulative risk.

97 3.2 Optimal dual alignment

98 Under convexity assumptions the constrained problem is equivalent to maximising the unconstrained
99 objective $R - \lambda S$ for a unique optimal dual λ Huang et al. [2024b]. Computing policies for $\lambda = 0$
100 (reward-only) and $\lambda \rightarrow \infty$ (safety-only) yields anchors θ_R and θ_S that bound the Pareto frontier.

101 3.3 VISAGE safety basin

102 The safety landscape exhibits a basin: for ℓ_2 distance $d(\theta, \theta_{safe}) \leq \delta$ the model’s refusal behaviour
103 remains stable; beyond δ safety collapses sharply Peng et al. [2024]. Retaining basin proximity is
104 therefore essential during fine-tuning and inference.

105 3.4 Refusal direction and low-rank control

106 A single activation direction mediates refusal across many chat models Ardit et al. [2024]. This
107 motivates our Safety Vector Field, which generalises to four orthogonal directions per layer and is
108 updated online.

109 3.5 Low-rank adaptation and threats

110 Parameter-efficient fine-tuning techniques such as LoRA, AdaLoRA, and DoRA insert rank- r updates
111 into projection layers Liu et al. [2024], Zhang et al. [2023]. Attackers can ship malicious adapters
112 indistinguishable from benign ones, necessitating an adapter-quarantine mechanism.

113 4 Method

114 PRANCE comprises an offline dual-stage alignment followed by an online enforcement engine.

115 4.1 Reward-safety dualisation

116 We run PeCAN on PKU-SafeRLHF-30k to learn a smooth convex dual and obtain $\lambda \approx 1.05$. Two
117 Direct Preference Optimisation (DPO) runs produce reward-optimised weights θ_R and risk-minimised
118 weights θ_S .

119 4.2 Lazy safety fusion

120 We form θ_0 by interpolating θ_R and θ_S with coefficient $\alpha = 0.35$ while minimising $L =$
121 $\text{KL}(\pi_\alpha \| \pi_R) + \text{KL}(\pi_\alpha \| \pi_S) + \mu \|\theta_\alpha - \theta_{safe}\|^2$ with $\mu = 0.05$, replicating Lisa’s proximal regu-
122 lariser Huang et al. [2024a]. This yields $d(\theta_0, \theta_{safe}) < 0.1$, comfortably inside the VISAGE basin.

123 4.3 Online safety engine

124 During generation each transformer block l stores four orthogonal vectors $v_{l,k}$. Given hidden state
 125 h_l we compute a layer-risk logit $r_l = \|\text{Proj}_V(h_l)\|_2$. SVF vectors are meta-updated every 256
 126 generated tokens using a replay-buffer objective that maximises mutual information between r_l and
 127 future GPT-4o risk labels while enforcing rank-4.

128 4.4 Hierarchical monitors

129 (1) A token-level classifier outputs $p_{tok}(\text{risk})$. (2) A lightweight ViT-T encodes sliding 64-token win-
 130 dows, producing $p_{seg}(\text{risk})$. (3) A trajectory LSTM consumes $(r_l, p_{tok}, p_{seg}, \text{decoding parameters})$
 131 and outputs Q_{help} and Q_{risk} .

132 4.5 Dual-policy controller

133 At each step the controller selects $a \in \{\text{pass}, \text{replace}, \text{truncate}, \text{refuse}\}$ that maximises $Q_{help} - \lambda \cdot$
 134 Q_{risk} . Replace swaps the candidate token with the safest top- k alternative; truncate ends the reply.

135 4.6 Robustness augmentations

136 Training randomly applies weight rotations, rank-5 ablations, back-door fusions, and one-shot harmful
 137 fine-tunes. A VISAGE penalty $\varepsilon \|\theta - \theta_{safe}\|^2$ discourages basin escape.

138 4.7 Adapter quarantine

139 On loading external adapters we compute the maximum cosine between their update directions and
 140 any SVF or refusal direction. If this exceeds $\tau = 0.48$ the adapter is sandboxed and λ is temporarily
 increased.

Algorithm 1 Online SVF meta-update with replay

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1: Initialise SVF directions  $\{v_{l,k}\}$  orthogonal per layer; replay buffer  $\mathcal{B} \leftarrow \emptyset$ 
2: for each generation step  $t$  do
3:   Obtain hidden states  $\{h_l^{(t)}\}$  and compute  $r_l^{(t)} \leftarrow \|\text{Proj}_V(h_l^{(t)})\|_2$ 
4:   Append  $(h^{(t)}, r^{(t)}, y^{(t+\Delta)})$  to  $\mathcal{B}$  where  $y$  are delayed risk labels
5:   if  $t \bmod 256 = 0$  then
6:     Sample minibatch  $\mathcal{M} \subset \mathcal{B}$ 
7:     Maximise  $I(r; y)$  via contrastive loss  $\mathcal{L}_{\text{MI}}$  on  $\mathcal{M}$ 
8:     Enforce rank-4 and orthogonality:  $V \leftarrow \text{orthonormalise}(V)$  per layer
9:     Update  $\{v_{l,k}\}$  by AdamW on  $\mathcal{L}_{\text{MI}} + \gamma \|V^\top V - I\|_F^2$ 
10:   end if
11: end for

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Algorithm 2 Dual-policy decoding control

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1: Given token proposal  $x_t$ , compute monitors  $p_{tok}, p_{seg}$  and features  $z_t$ 
2: Predict  $Q_{help}(z_t), Q_{risk}(z_t)$  via trajectory LSTM
3: Select  $a \leftarrow \arg \max_{a \in \{\text{pass}, \text{replace}, \text{truncate}, \text{refuse}\}} (Q_{help}(a) - \lambda Q_{risk}(a))$ 
4: if  $a = \text{pass}$  then
5:   Emit  $x_t$ 
6: else if  $a = \text{replace}$  then
7:   Emit  $\arg \min_{x \in \text{top-}k} Q_{risk}(x)$ 
8: else if  $a = \text{truncate}$  then
9:   End response
10: else
11:   Emit refusal token and end
12: end if

```

142 5 Experimental Setup

143 5.1 Hardware and framework

144 All experiments run on a single NVIDIA A100-40 GB GPU using PyTorch 2.2, vLLM, and flash-
145 attention. Weights are fp16; activations bf16.

146 5.2 Model and training

147 Base model: Llama-3-Instruct-2 (3 B). Dual-stage alignment: one PeCAN epoch plus DPO on
148 PKU-SafeRLHF-30k, batch 512 prompt tokens, learning rate 2×10^{-5} , LoRA rank 16. Lisa fusion
149 uses $\lambda = 0.3$ and $\mu = 0.05$. Enforcement modules (SVF, Q-nets, ViT-T, LSTM) train for two hours
150 with AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$) and learning rate 1×10^{-4} .

151 5.3 Data

152 Alignment: PKU-SafeRLHF-30k augmented with Anthropic Constitutional and BeaverTails safe
153 prompts. Evaluation: AdvBench-500 Chao et al. [2024], HarmBench-159, JailbreakBench-100, plus
154 Spanish and Arabic translations. Weight attacks include Universal-Backdoor fusions, Lisa harmful
155 fine-tuning, and random rank-5 ablations.

156 5.4 Metrics

157 Safety: attack-success rate (ASR) scored by GPT-4o rubric. Robustness: Δ ASR under each attack.
158 VISAGE depth: ℓ_2 distance to safety-basin centre. Utility: MT-Bench score, MMLU accuracy,
159 GSM8K exact match. Efficiency: median latency over 256 prompts and peak VRAM.

160 5.5 Baselines

161 (i) Constitutional-AI-3B retrained on identical data, (ii) SmoothLLM-3B, (iii) TCM-3B. All share the
162 3 B parameter budget and identical LoRA recipe but omit PRANCE modules.

163 5.6 Experiments

164 1) Implementation-Trace Alignment Test (ITAT) validates module execution via autograd traces and
165 a micro ablation. 2) Scaled End-to-End Robustness Evaluation (SERIE) sweeps a 4×4 decoding grid
166 across three weight states for 1 536 prompt points per model. 3) Seven-Day Drift & LoRA Infection
167 Simulation (SDDL) mimics continual deployment with daily reward fine-tunes and random adapter
168 injections.

169 6 Results

170 6.1 Implementation-Trace Alignment Test

171 All six PRANCE modules appear in the autograd trace, and WandB logs are populated. Disabling
172 SVF raises ASR on a 64-prompt AdvBench slice from 2.8 % to 11.5 % ($\Delta = +8.7$ pp), surpassing
173 the 5 pp pass criterion. Wall-clock time is 14 min; peak VRAM 12 GB.

174 6.2 Scaled End-to-End Robustness Evaluation

175 Clean ASR: PRANCE $2.4 \% \pm 0.9$ (AdvBench-128), $3.9 \% \pm 1.7$ (JailbreakBench-40), $1.5 \% \pm 0.8$
176 (HarmBench-40). Constitutional-AI records 21.8 %, 18.6 %, 12.9 %; SmoothLLM 9-12 %. Malicious
177 LoRA merge increases PRANCE ASR by +2.2 pp, versus +19.4 pp baseline. Toxic Lisa-900 fine-tune
178 adds +2.7 pp for PRANCE versus +18.1 pp baseline. Changing decoding from ($T = 0.2, p = 0.95$)
179 to ($T = 0.7, p = 0.3$) raises PRANCE ASR by ≤ 1.8 pp. Utility cost is minimal: MT-Bench-Lite
180 -0.8 %, MMLU -0.4 pp, GSM8K unchanged. Latency overhead 11.3 %; peak VRAM 10.9 GB -
181 both within deployment budgets.

182 6.3 Hyper-parameter sensitivity

183 ASR is convex in λ with a plateau between 1.0-1.3. SVF update period 256 trades latency and safety
184 best: updating every 64 tokens adds 4 % latency; every 1 024 tokens increases ASR by 3 pp.

185 6.4 Seven-Day Drift & LoRA Infection Simulation

186 PRANCE-online ASR stays 2.9-4.7 % (AUC $\approx$ 26). PRANCE-static (frozen SVF) drifts to 16.1
187 % by day 7; SmoothLLM exceeds 15 % by day 4. VISAGE depth for PRANCE-online remains
188 0.08 ± 0.02 , whereas PRANCE-static breaches 0.15 on day 5. Adapter quarantine detects 22 / 24
189 malicious inserts (recall 0.92) with two false positives. Daily latency overhead 12-14 %.

190 6.5 Ablations

191 Removing the trajectory LSTM increases ASR by 3.4 pp; omitting ViT-T adds 2.1 pp; disabling the
192 VISAGE penalty raises weight-attack Δ ASR to 6.8 pp, confirming each component’s contribution.

193 6.6 Limitations

194 PRANCE depends on GPT-4o labels for SVF training; very long conversations moderately increase
195 compute due to frequent updates. Extremely large harmful fine-tunes ($>1\,000$ steps) can eventually
196 escape the safety basin, albeit more slowly than baselines.

197 7 Conclusion

198 We introduced PRANCE, a resource-efficient defence that couples optimal dual alignment, proximal
199 safety fusion, and adaptive low-rank enforcement to counter prompt-, decoding-, and weight-
200 space jailbreaks. Across implementation-trace tests, large-scale robustness grids, and a week-long
201 continual-learning simulation, PRANCE cuts attack success by an order of magnitude relative to a
202 strong Constitutional-AI baseline, maintains helpfulness within 1 %, and meets edge-deployment
203 budgets. Our results highlight that maintaining a dynamically updated safety sub-space, enforced
204 by a principled dual-policy controller, can deliver robust, helpful, and harmless LLMs at modest
205 cost. Future work will extend SVFs to mixture-of-experts architectures, incorporate pluralistic value
206 alignment Sorensen et al. [2024], and pursue certified guarantees via SafeRL theory Ji et al. [2023].
207 Stress-testing against emerging attacks such as COLD-Attack and Charmer remains an open priority.

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